CLIVORA

QUANTIC MSSE · CAPSTONE PRESENTATION

Freelance work. One connected system.

The AI-assisted operating system where independent professionals find work, run the delivery and get paid — in one place.
Solo Capstone · three sprints · web and Android, one shared record.

WEB · LIVE AT CLIVORA.IO

ANDROID · GOOGLE PLAY



Usman Ghias — Product Owner, Scrum Master and sole engineer · CODCrafters
github.com/UsmanGhias/clivora

HOW THIS SESSION RUNS

Agenda

Product first, then the engineering behind it, then the agile record, then a live demo.

01

The problem

Why independent work fragments across tools

02

The product

Web and Android walkthrough with real screens

03

Engineering

Architecture, patterns, security, AI, tests, CI/CD

04

Agile process

Baseline vs Capstone, three sprints, the board

05

Live demo

Job → accept → project → invoice → paid

06

Close

Results, submission links, next roadmap

TEAM, ROLES AND HONEST SCOPE

One person, three roles, and a clear line around the grading

ROLE

Product Owner

Wrote and ranked the user stories, accepted or rejected each sprint increment, owned the release call.

ROLE

Scrum Master

Ran sprint planning, kept the GitHub board honest, enforced the definition of done and the timebox.

ROLE

Engineer

All Flutter, Next.js, SQL, CI and release work. Every commit in the repo is attributable to me.

Scope statement

CLIVORA already existed before this Capstone. The baseline product — CRM, invoicing, Connect, offline sync, billing — shipped earlier and I do not present it as sprint work.
The three sprints being graded are a new increment: test and CI hardening, turning Ask AI into a real retrieval-augmented system, and one reliable hire-to-invoice demo loop.

01

SECTION ONE

The problem

Not another marketplace. A connected way to move independent work forward.



CLIVORA PREMIUM CAMPAIGN V2

SECTION 01 · THE PROBLEM

Independent work is not hard because of the work. It is hard because of the seams.

01

Discovery is walled off

Marketplaces own the client relationship and take a cut of every deal.

02

Delivery is scattered

Chat in one app, tasks in another, files in a third, hours in a spreadsheet.

03

Money leaks

Invoices are rebuilt by hand, follow-ups are forgotten, revenue is invisible.

04

Clients are left guessing

No shared place to see progress, approve work or pay against a record.

2 sides

One product has to serve the freelancer and the client, or the seam stays

$0

Platform fee CLIVORA takes on a negotiated Connect deal

Offline

Mobile work happens on bad networks — the app cannot assume a connection

1 record

Brief, delivery and invoice must resolve to one auditable trail

USERS AND THE ALTERNATIVES

Who it is for, and why the existing options do not fit

USER 01

Solo freelancer

Needs one place for clients, projects, hours and invoices — on a phone, often offline.

USER 02

Independent team / agency

Needs shared projects, task assignment, roles and a single revenue picture.

USER 03

The client

Needs to see progress, approve work, message once and pay against a real invoice.

WHY NOT JUST USE WHAT EXISTS

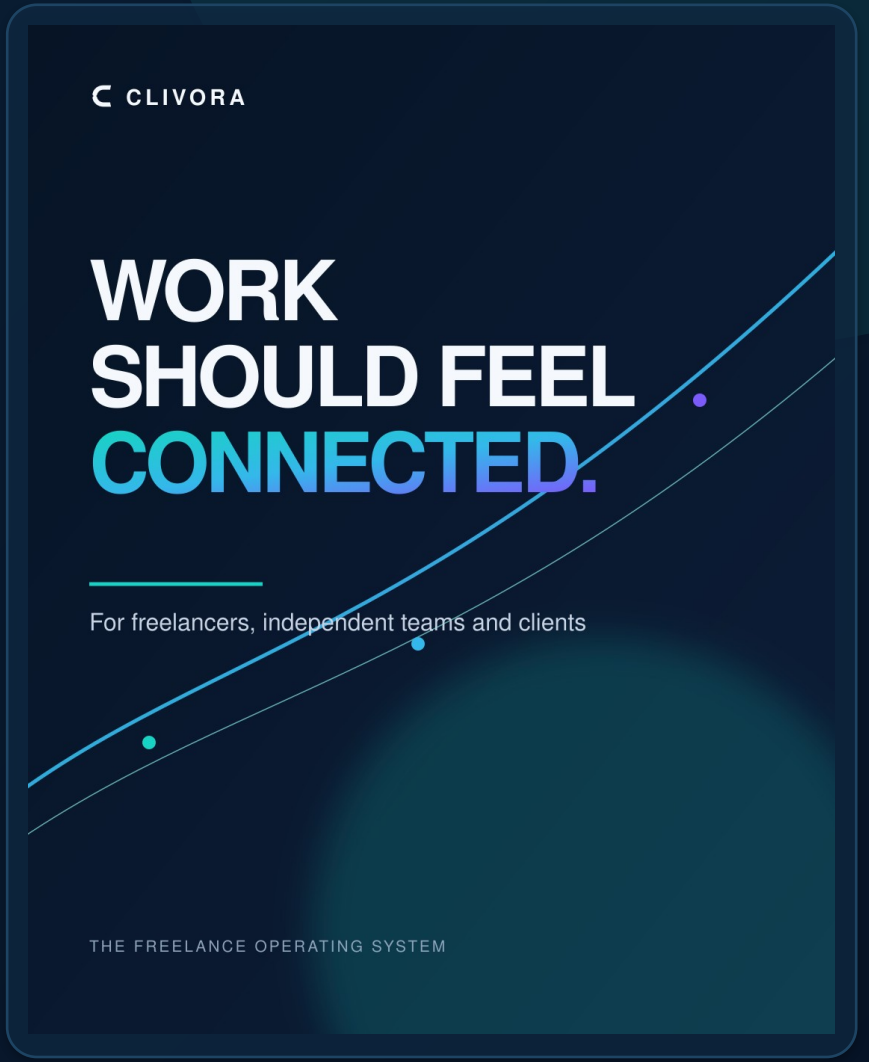
Freelance marketplaces	Own the client relationship and charge a percentage of each deal
Generic project tools	No invoicing, no client portal, no concept of a freelance plan
Accounting apps	Invoice-only; nothing about delivery, tasks or the client conversation
Spreadsheets + chat	Zero automation, no offline guarantees, no shared client view

02

SECTION TWO

The product

Work should feel connected — for freelancers, independent teams and clients.

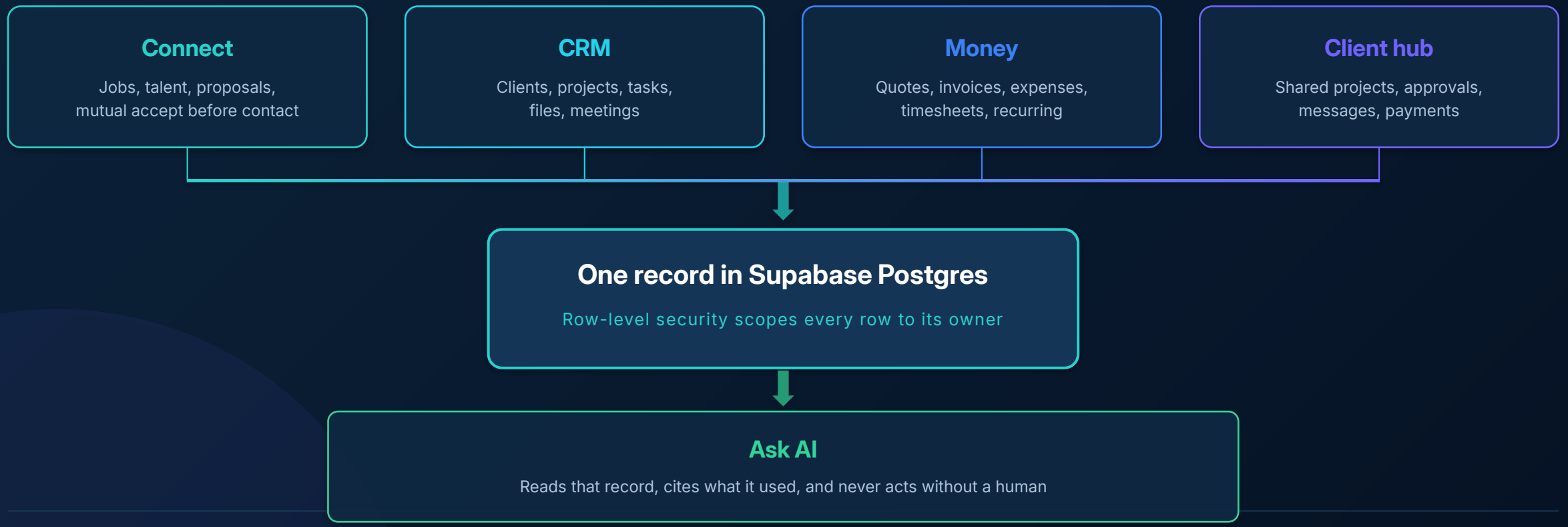


CLIVORA PREMIUM CAMPAIGN V2

SECTION 02 · THE PRODUCT

One workspace, five connected surfaces

Everything below writes to the same Postgres record, so nothing has to be re-entered.



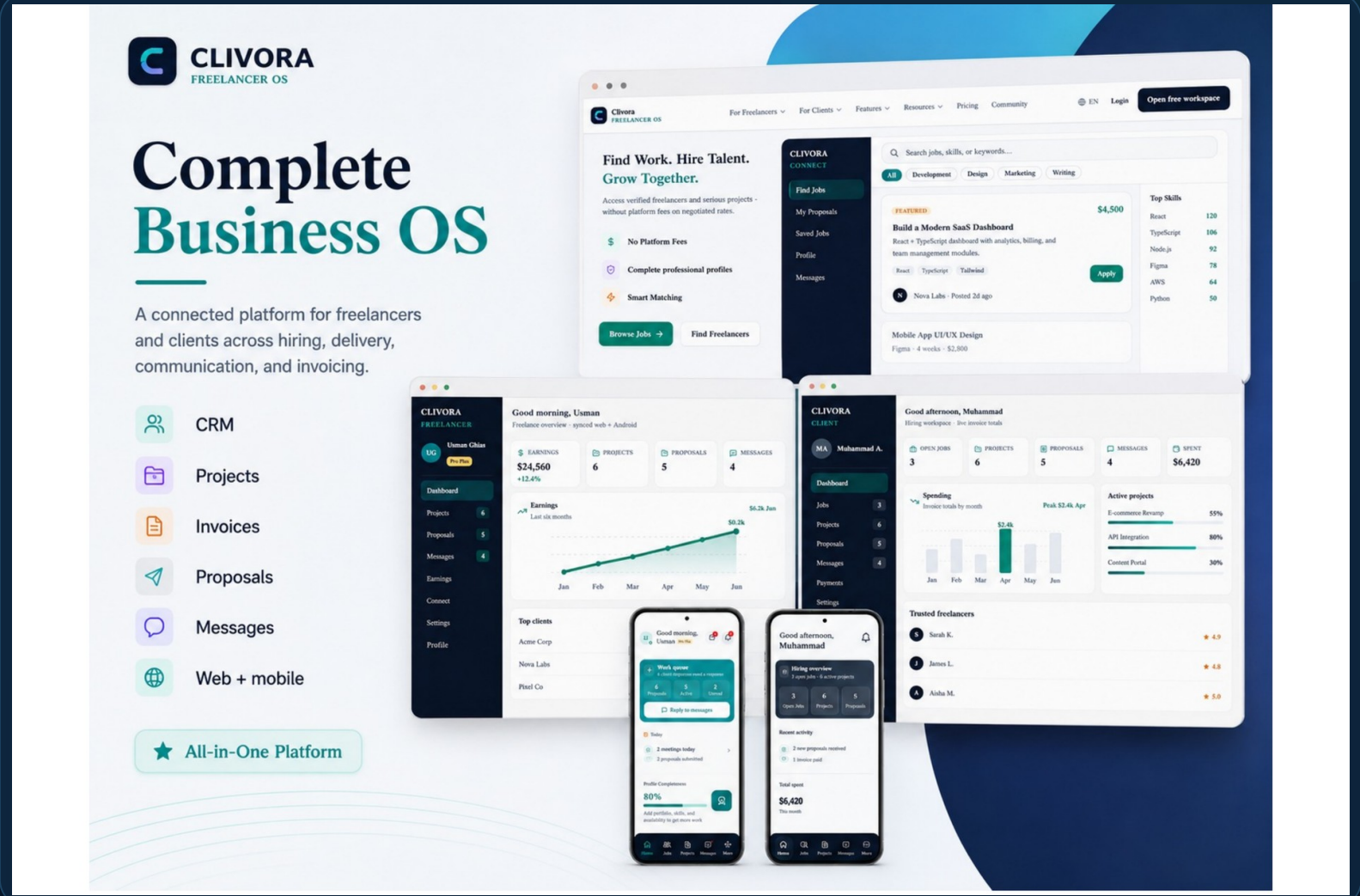
WEB · LIVE AT CLIVORA.IO

The freelancer workspace

The web app is a Next.js 15 App Router product deployed as a Node process on Hostinger. Server Components load the profile and gate every route before the browser renders anything.

This is the surface a freelancer opens in the morning: what needs attention, what is in flight, and what has not been paid.

- Server-gated routes, no client-side auth
- Same Postgres rows as the Android app
- Public invoice share links at /i/[token]

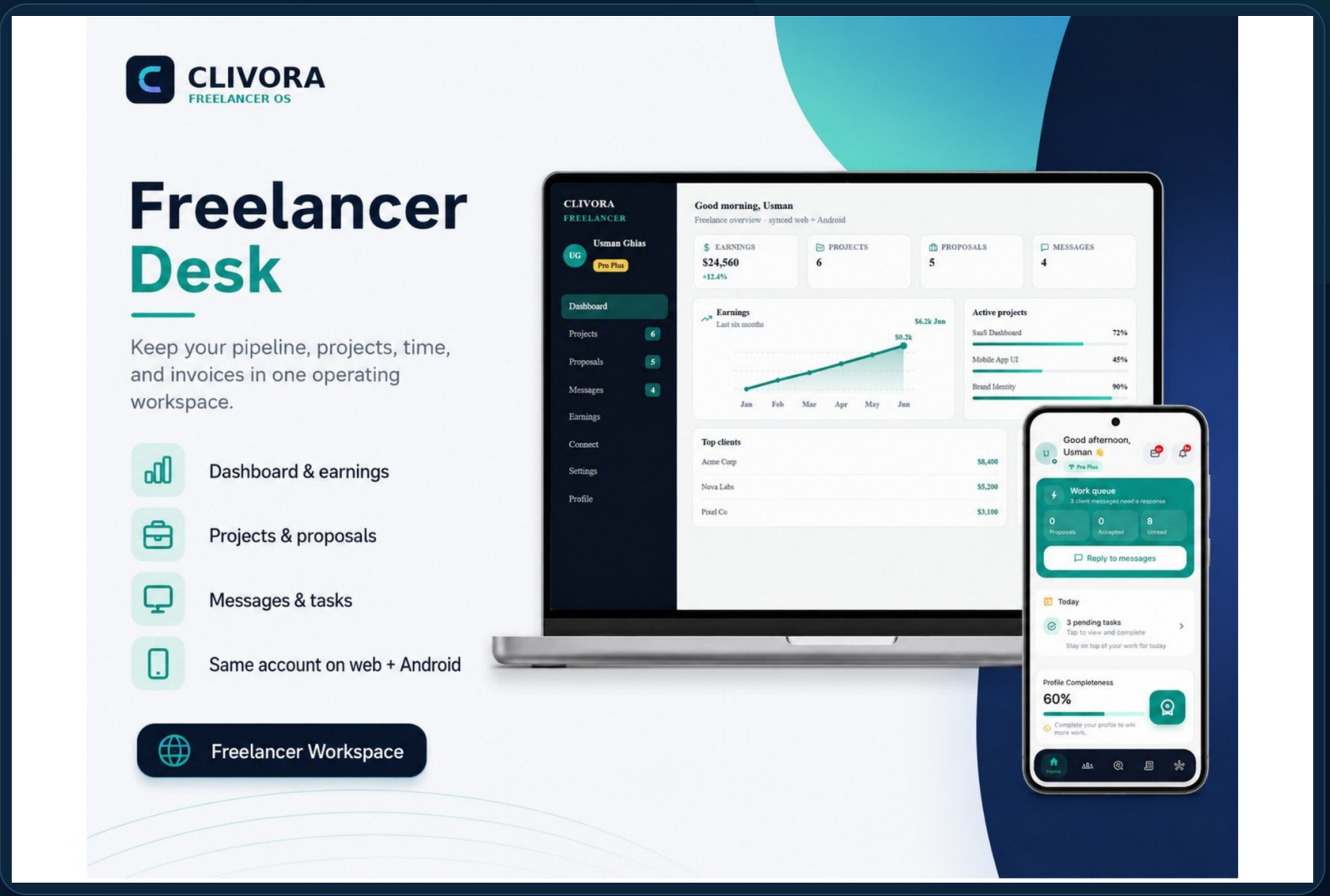


THE DAILY LOOP

From daily priority to final invoice

Six steps cover almost everything a freelancer does in a day. CLIVORA is designed so each step hands off to the next without an export, a copy-paste or a second tool.

- **Start the day**
- **Organise the work**
- **Keep clients close**
- **Discover opportunities**
- **Track the time**
- **Send the invoice**



CLIVORA CONNECT · NOT ANOTHER MAR

Find the work. Keep the relationship.

Connect is a private marketplace. Contact details stay hidden until both sides accept, and CLIVORA takes no percentage of the negotiated deal.

Mutual accept

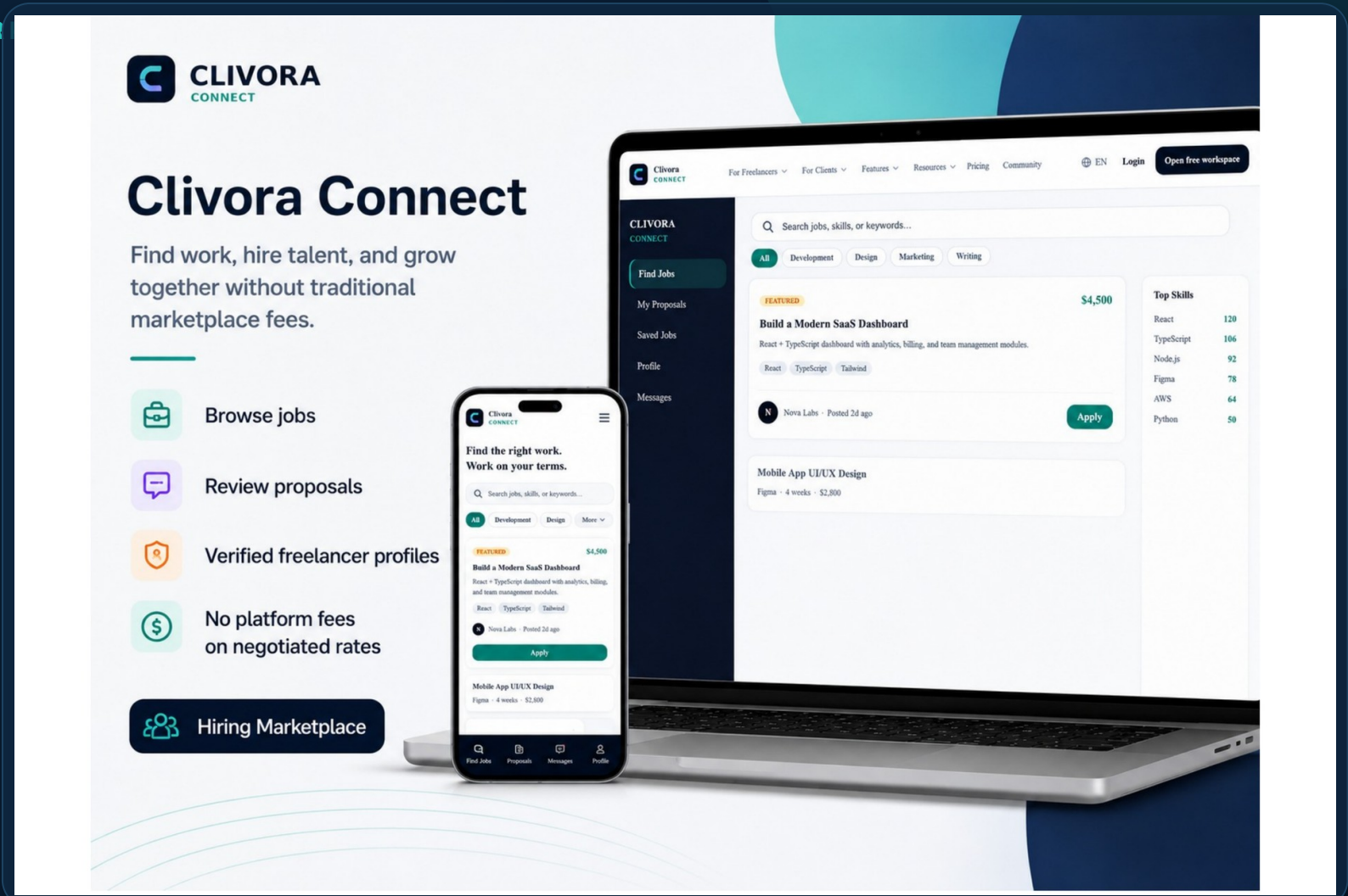
Emails are revealed only after both parties agree

$0 platform fee

Revenue comes from subscriptions, not from your deal

Pro Plus to act

Browsing is free; posting and contacting is an entitlement

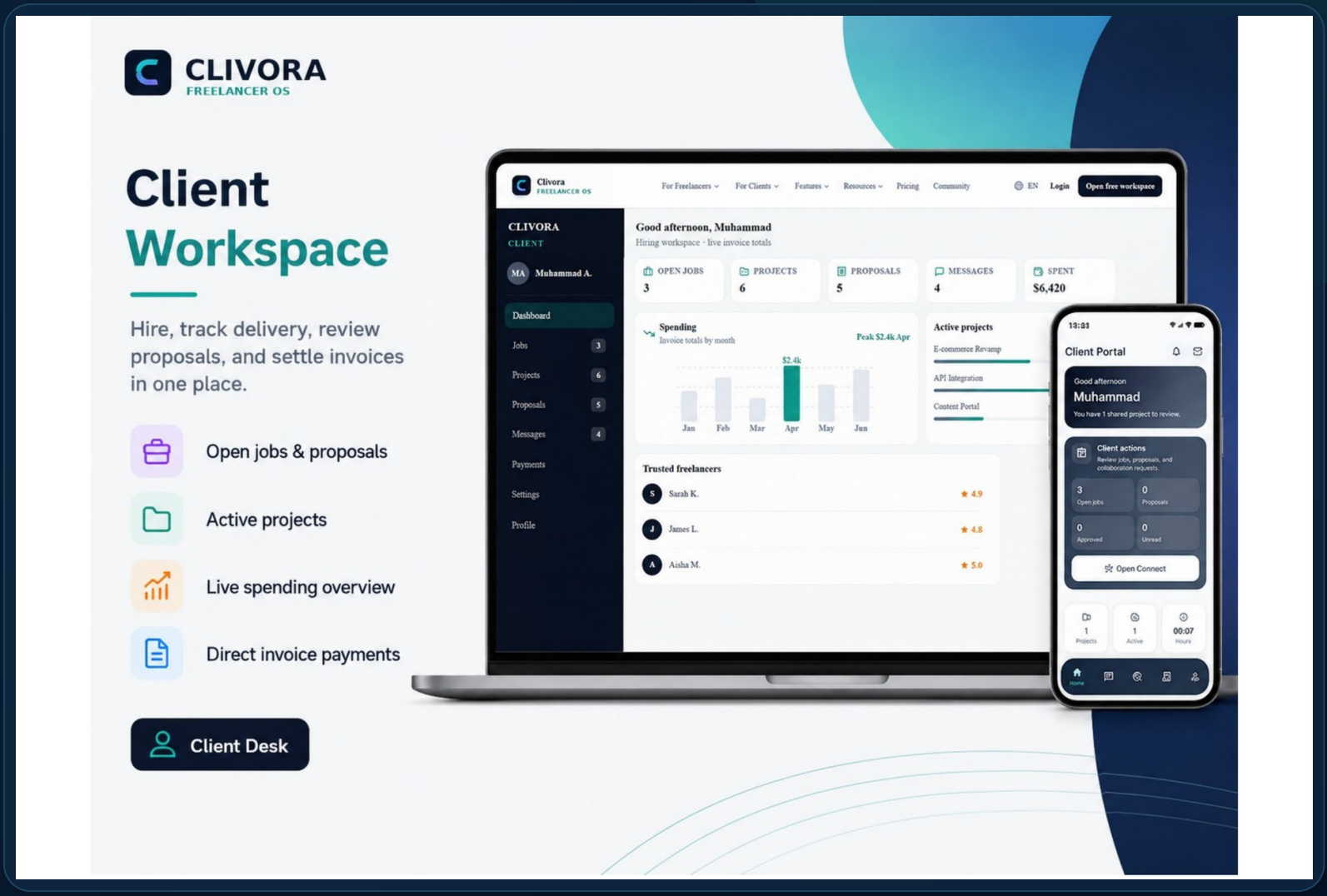


BUILT FOR CLIENTS

The client is a first-class user

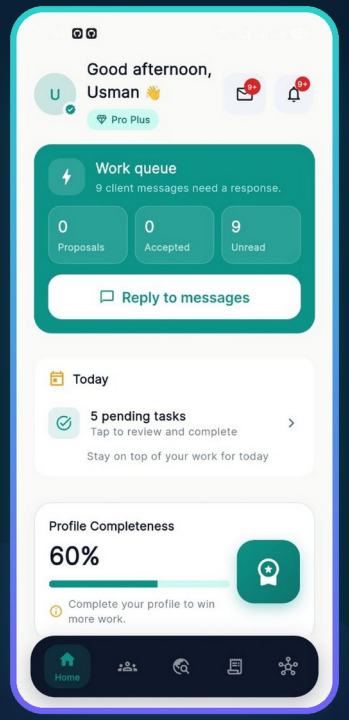
Clients get a real account, not a share link. They see the projects that were explicitly shared with them, approve work, message once, and pay against an invoice that already exists in the system.

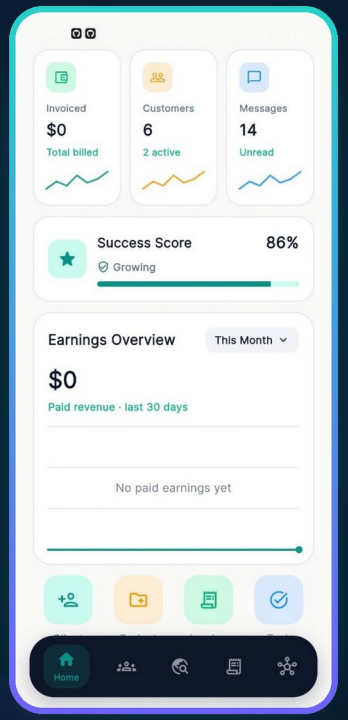
- Shared projects with live progress
- Client actions queue: jobs, proposals, approvals
- Invoices and contracts in one hub
- One thread per relationship

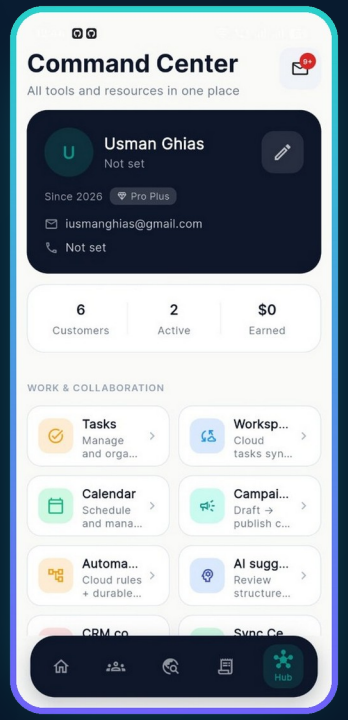


ANDROID · FLUTTER · GOOGLE PLAY

The freelancer's phone, unretouched







01 Work queue first

Unanswered client messages, accepted work and today's tasks before anything else.

02 The business at a glance

Invoiced total, active customers, unread messages, success score and 30-day revenue.

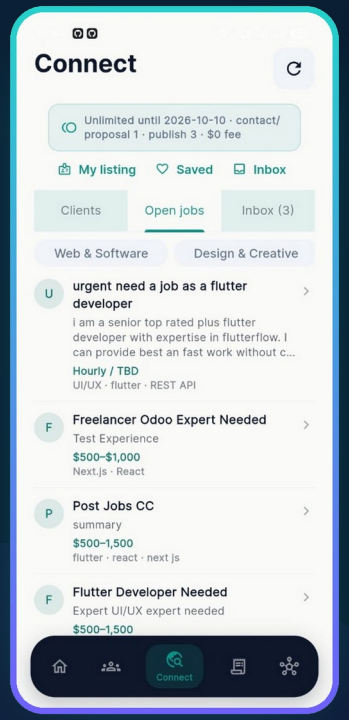
03 Command Center

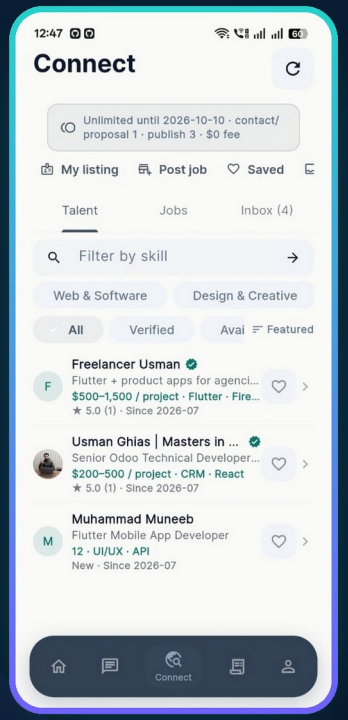
Every module reachable in two taps — tasks, calendar, campaigns, automations, AI.

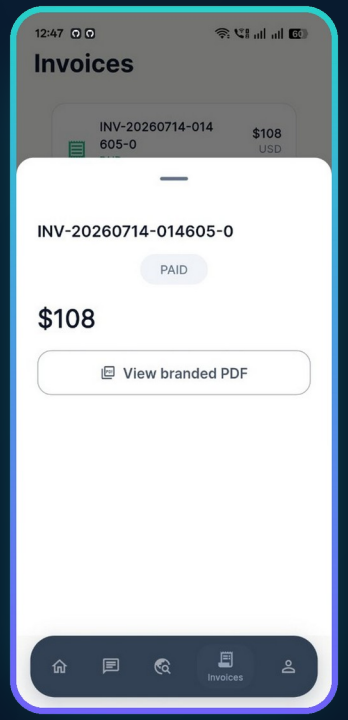
UNRETOUCHED SCREENSHOTS FROM THE RELEASED ANDROID BUILD

ANDROID · CONNECT AND MONEY

Discovery and getting paid







UNRETOUCHED SCREENSHOTS FROM THE RELEASED ANDROID BUILD

01 **Open jobs**

Live Connect listings with budget bands and skills. Browsing is free on every plan.

02 **Verified talent**

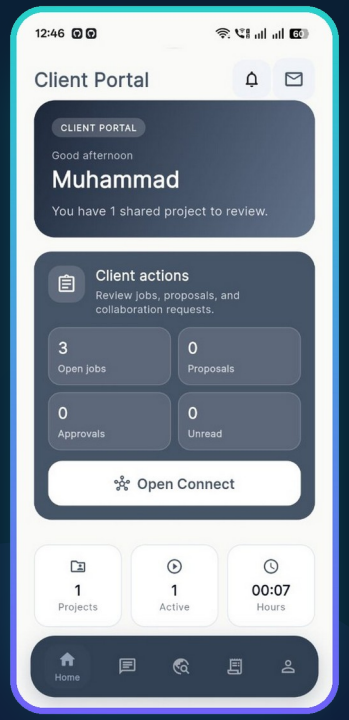
The other side of the same marketplace — verified profiles, rates and history.

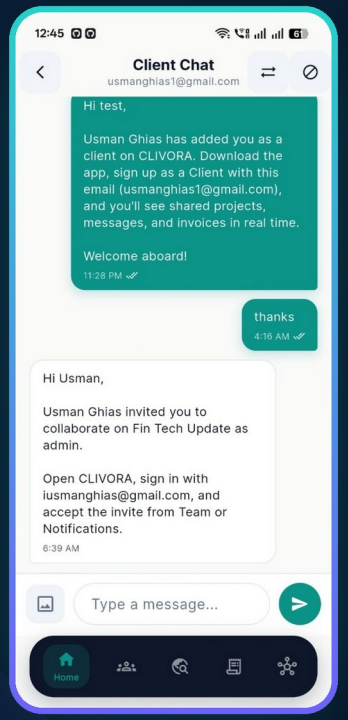
03 **Paid, with a branded PDF**

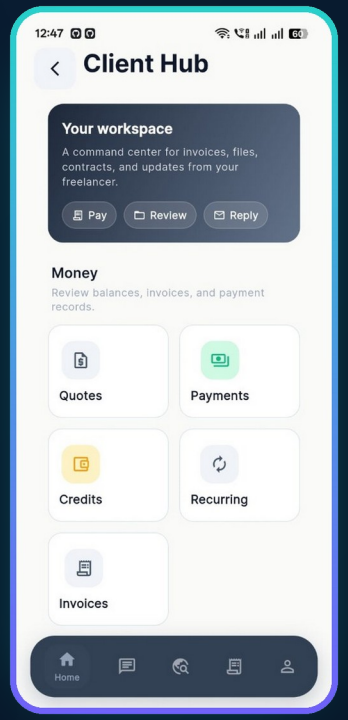
A real invoice moved to PAID, with a generated branded PDF the client can keep.

ANDROID · THE CLIENT SIDE

The client's phone is a different app







UNRETOUCHED SCREENSHOTS FROM THE RELEASED ANDROID BUILD

01 Client portal

A client signs in and sees their action queue and only the projects shared with them.

02 One thread per relationship

Invitations, updates and replies in a single conversation, synced to the web app.

03 Client hub

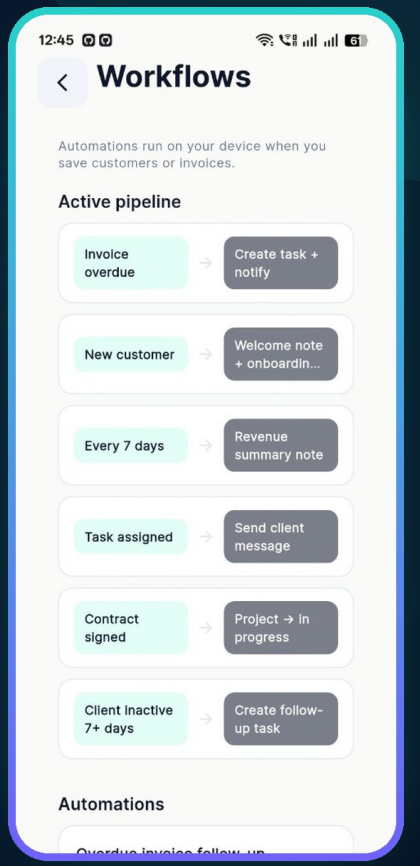
Quotes, payments, credits, recurring and invoices — the money view for the buyer.

AUTOMATION ENGINE

Rules that run on the device

Automations fire locally when a customer or invoice is saved, so follow-ups happen even when the network does not.

Invoice overdue	Create task + notify	**New customer**	Welcome note + onboarding
Every 7 days	Revenue summary note	**Task assigned**	Send client message
Contract signed	Project → in progress	**Client inactive 7+ days**	Create follow-up task



ONE ACCOUNT, TWO CLIENTS

The same product, not a port

Shared source of truth

Both clients read and write the same Postgres rows. There is no sync layer between two databases, because there is only one.

Shared business rules

plan-gates.ts on the web and PlanLimits in Flutter are mirrored, and both are unit-tested.

Different strengths

Web gets server-rendered route gating. Android gets offline-first storage and on-device automation.

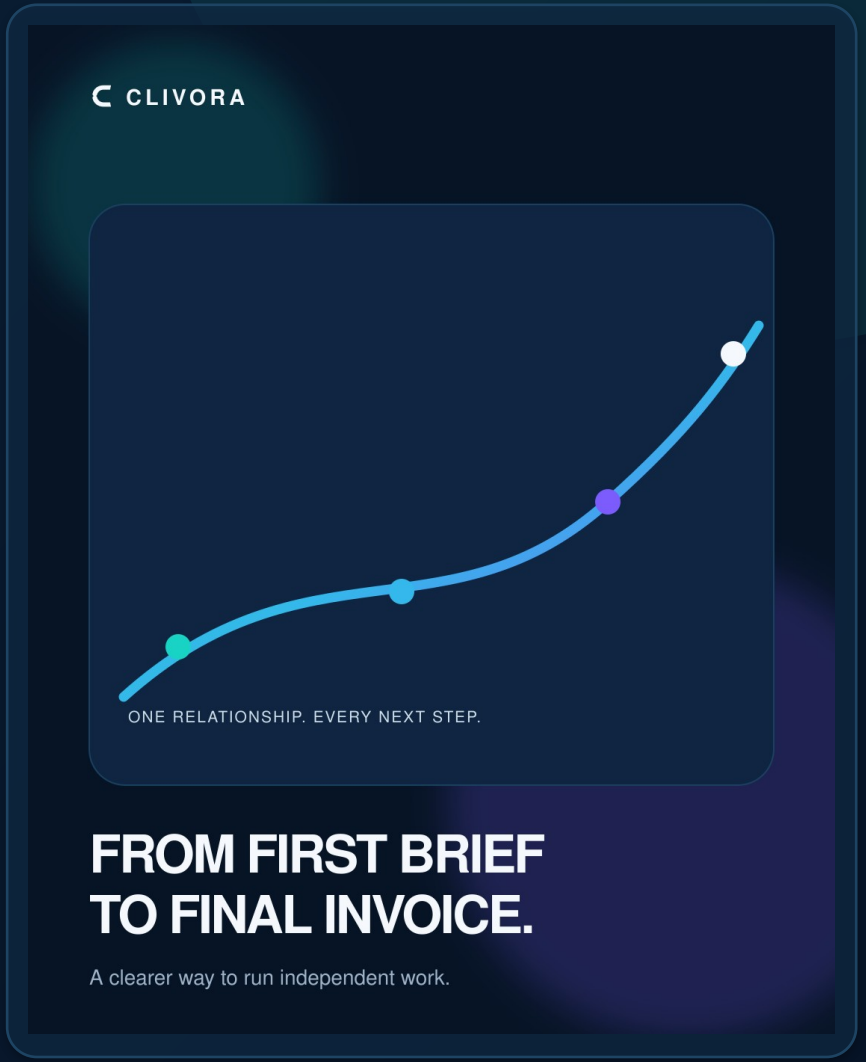


03

SECTION THREE

Engineering

From first brief to final invoice — one auditable record behind every screen.

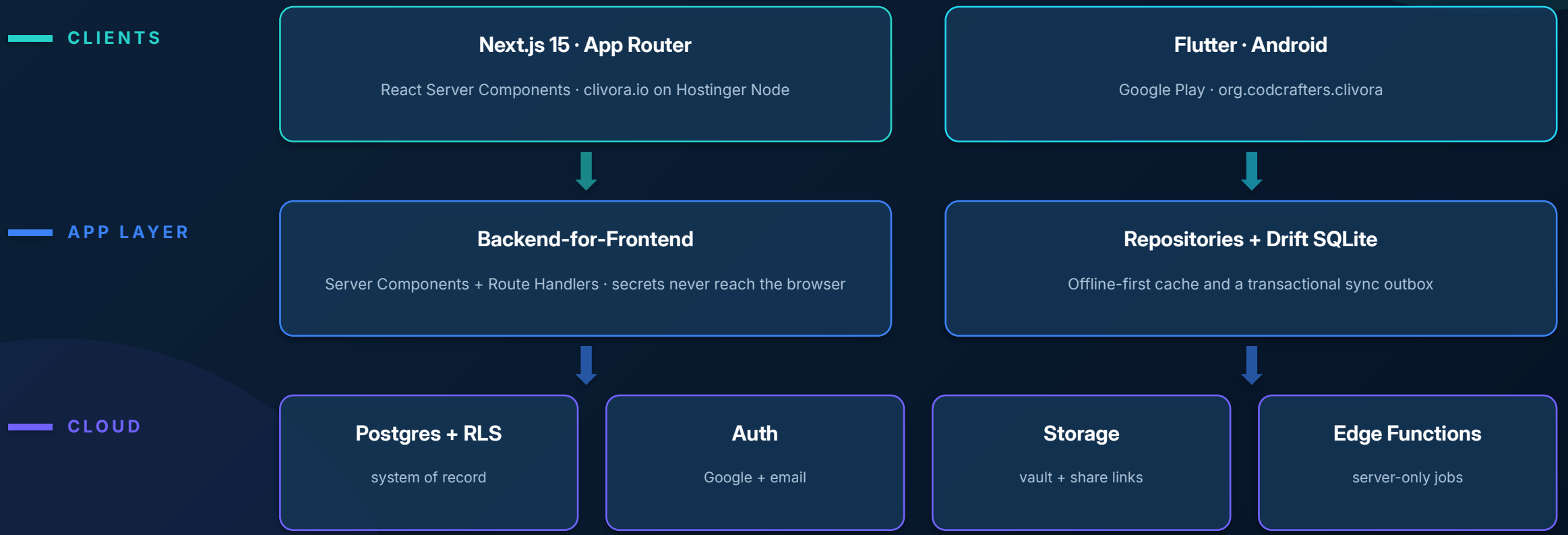


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SECTION 03 · ENGINEERING

System architecture

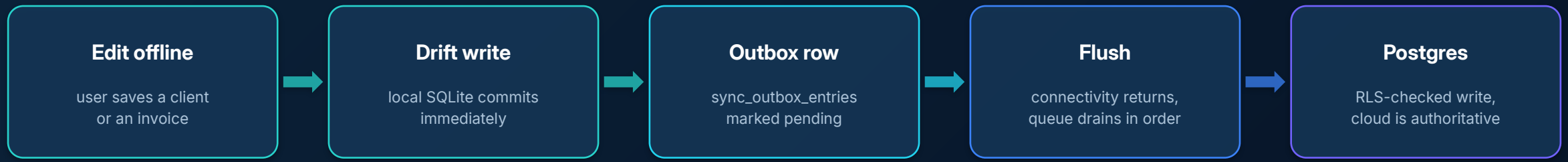
Two untrusted clients, one trusted database. Authorisation lives in Postgres, not in the UI.



PATTERN · TRANSACTIONAL OUTBOX ON A MOBILE CLIENT

Offline-first sync, and what happens when sync fails

Freelancers work on bad networks. The Android app never blocks a write on connectivity.



Failure path is explicit

Unknown operation kinds are marked permanently_failed and kept for diagnosis rather than silently dropped. Last-write-wins collisions write a conflict record.

Why this pattern

It is the transactional outbox from distributed systems, adapted to a phone. Tested with in-memory Drift in test/sync_outbox_service_test.dart.

PATTERN · AUTHORISATION IN THE DATA LAYER

Security model: the database is the last word

Both a browser bundle and an APK can be read by an attacker, so neither can be the gate.

Client
FLUTTER APK / BROWSER BUNDLE

Assumed hostile. Hiding UI is a UX choice, never a control.

Server
ROUTE HANDLERS + SERVER COMPONENTS

Cookie sessions via @supabase/ssr. LLM keys and service role stay here.

Database
POSTGRES ROW-LEVEL SECURITY

Every table policy-scoped to the owner. Triggers protect role, account type and plan flags from self-elevation.

Concrete controls

- Connect emails hidden until mutual accept
- Vault share links resolve owner-scoped files, never a raw caller path
- Privileged profile columns guarded by DB triggers
- No service-role key in any shipped client
- Secrets, keystores and .env gitignored

DESIGN DECISIONS

Patterns used, and why each one was chosen

Feature-first modules

lib/features/* and App Router areas.
Sprint slices map to folders, so invoices never bleed into Connect.

Repository pattern

CloudCrmRepository, CloudInvoiceRepository, web/src/lib/crm.
One place for queries, mapping and entitlement checks.

Transactional outbox

Offline writes queue and drain in order.
Chosen because mobile connectivity cannot be a precondition.

Backend-for-Frontend

Server Components and route handlers hold the session.
Keeps secrets out of the browser bundle and the APK.

Policy objects

plan-gates.ts and Flutter PlanLimits.
Product rules become testable functions, not UI copy.

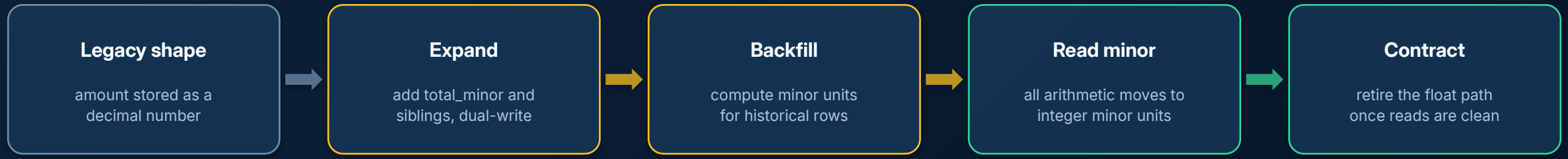
Expand / contract

Money stored as major units and integer minor units.
Migration without a flag-day cutover.

CORRECTNESS WHERE IT HURTS MOST

Money is never a floating-point number

An invoice that is a cent wrong destroys trust faster than any missing feature.



Proven by tests, not by inspection

- web/src/lib/money.test.ts — conversion + dual-write fields
- Helpers are pure functions in web/src/lib/money.ts

Entitlements get the same treatment

- Free caps: 3 clients · 5 projects · 10 invoices per UTC month
- plan.test.ts and plan-gates.test.ts pin the boundaries

AI ENGINEERING

Ask AI: retrieval-augmented, grounded, and never autonomous

Sprint 2 turned a generic assistant into a system that answers from the user's own records.



Human-in-the-loop gate

Structured suggestions are feature-flagged and require review. auto_pay and auto_publish are forbidden — the model proposes, a person commits.

Fail open, honestly

No key or no index? Ask AI falls back to CRM aggregates and honest offline tips rather than inventing an answer.

Indexing is opt-in

POST /api/ai/index-crm. Live invoice and client saves were left unchanged, so the AI work could not regress the money path.

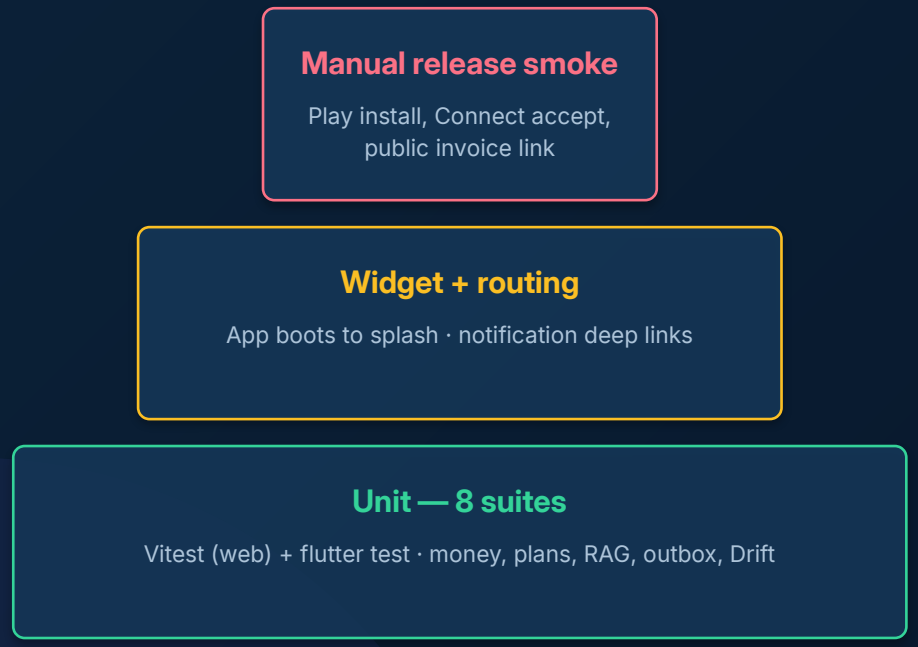
Evaluated, not vibes

A written eval set of questions and expected citations lives in docs/capstone/ASK_AI_EVAL.md.

VERIFICATION

Test strategy: cheap tests where the blast radius is largest

Eight automated suites run in CI on every push and pull request to main and develop.



Fast, deterministic, no device or cloud credentials needed

WHAT EACH SUITE PROVES

money.test.ts	Minor-unit conversion and invoice dual-write fields
plan-gates.test.ts	Free-plan caps and UTC-month invoice counting
crm-rag.test.ts	Chunking, ilike sanitisation, no invented amounts
plan.test.ts	Pro / Pro Plus / client detection
sync_outbox_service_test	Pending vs permanently_failed, payload migration
database_test.dart	Drift insert/read and per-user invoice counts
notification_deep_link	Notifications route to the right screen
widget_test.dart	App boots to the CLIVORA splash

DELIVERY

CI/CD: the same gates for the Capstone and for production

GitHub Actions on main and develop. A test that does not run in CI is not evidence.

CONTINUOUS INTEGRATION



CONTINUOUS DELIVERY



DEPLOYMENT RECOMMENDATION

Where this should run, and what it costs relatively

The handbook asks for a recommendation with relative cost. Mine is to stay on managed cloud — and the reason is architectural, not only financial.

OPTION	WHAT IT MEANS	RELATIVE COST	RECOMMENDATION
Managed cloud CHOSEN	Hostinger Node for Next.js · Supabase cloud · Google Play for Android distribution	Low to moderate. No Kubernetes bill, no ops headcount. Already live.	Recommended for the Capstone and early production.
Heavier public cloud DEFERRED	Same app on AWS or GCP — Cloud Run or ECS, RDS, S3, load balancers	Higher monthly cost and materially higher ops time at this traffic level.	Only once Hostinger or Supabase limits are actually reached.
On-premises NOT RECOMMENDED	Customer-hosted Postgres and Node, sideloaded or MDM-managed Android	High: hardware, TLS, backups, 24/7 ops, Play enterprise distribution.	Wrong fit — Connect is a two-sided network and needs a shared service.

04

SECTION FOUR

Agile process

Do the work. Own the momentum. Three sprints, planned and demonstrated.



CLIVORA PREMIUM CAMPAIGN V2

SECTION 04 · AGILE PROCESS

Baseline versus the Capstone increment

This is the same boundary from slide three, now with the evidence attached.

BASELINE — SHIPPED BEFORE THE CAPSTONE

- Freelancer and client accounts, Google sign-in
- CRM: clients, projects, invoices, expenses, quotes, timesheets
- Android offline CRM with the sync outbox
- Connect: jobs, talent, proposals, mutual accept
- Messaging, meetings, public invoice share pages
- Free / Pro / Pro Plus gates, Play Billing
- Admin console, feature flags, vault share links

CAPSTONE INCREMENT — WHAT QUANTIC GRADES

SPRINT 1

Grader-ready repo: docs, board, secrets hygiene, unit tests, npm test in CI

SPRINT 2

Ask AI becomes retrieval-augmented, with an eval set and a human-review gate

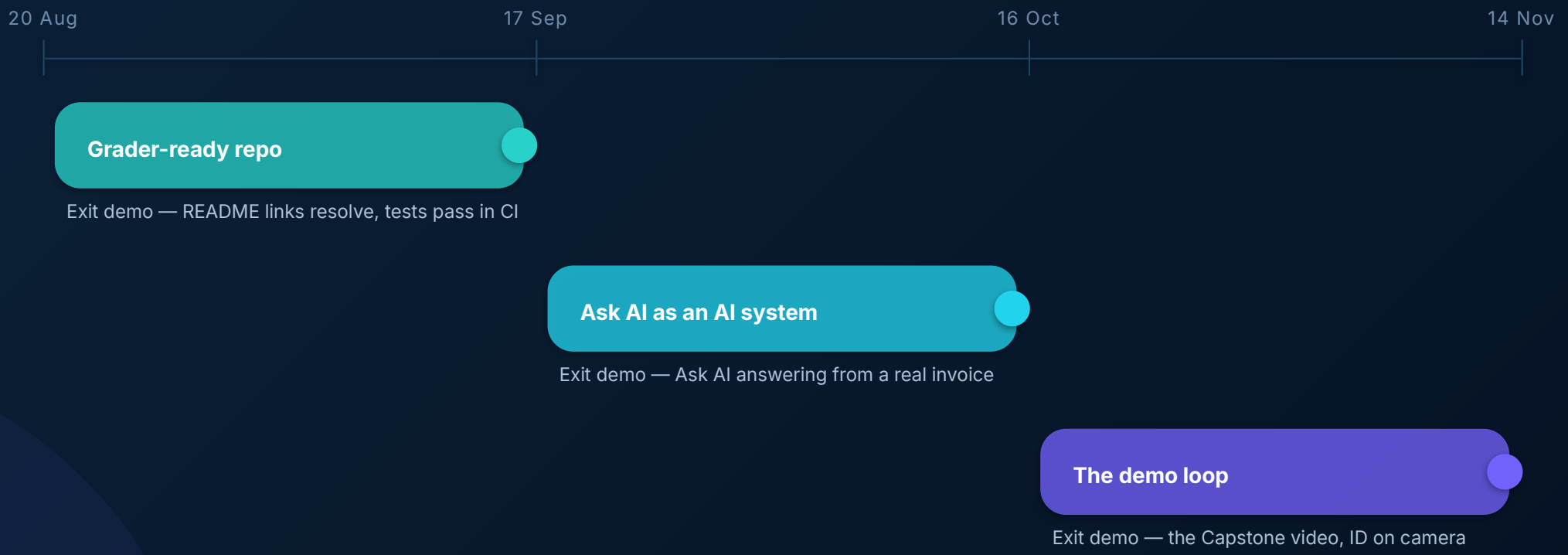
SPRINT 3

One reliable hire-to-invoice demo loop and this presentation

CADENCE

Three sprints, 20 August to 14 November 2026

Four weeks each, solo team, weekly timeboxed check-ins. Every sprint closes with a demo recorded for the Product Owner — accept or reject, then re-plan.



SPRINT DETAIL

What was planned, delivered and demonstrated

SPRINT 1 20 Aug – 17 Sep

Grader-ready engineering

- docs/capstone/ written: design and testing, stories, sprint plan
- GitHub Project and capstone-labelled issues
- Secrets, keystores and .env moved into .gitignore
- New unit suites: plan gates, money, outbox
- npm test wired into the CI workflow

SPRINT 2 18 Sep – 16 Oct

CLIVORA as an AI system

- Retrieval over crm_embeddings in the Ask AI route
- Opt-in indexing endpoint so saves stay untouched
- crm-rag unit tests: chunking, sanitisation, no invented amounts
- Written eval set in ASK_AI_EVAL.md
- Human-review gate: auto_pay and auto_publish forbidden

SPRINT 3 17 Oct – 14 Nov

One loop, end to end

- Scripted hire → project → timesheet → invoice → paid
- Client-side verification on a second real account
- Demo script committed so the run is repeatable
- This deck, speaker notes and the recorded walkthrough
- Final submission package for Quantic

WORKING AGREEMENT

The board and the definition of done

DEFINITION OF DONE — EVERY STORY

- Code on main, or in a pull request into main
- Tests added or updated whenever the behaviour is testable
- Board card moved to Done — not left in progress
- No secrets committed, verified before push
- Baseline features never relabelled as new Capstone work

COLLABORATIVE TOOLING

GitHub Issues + Project
Labels capstone, sprint-1/2/3. Public board for graders.

GitHub Actions
CI on every push; tag-driven Android release jobs.

Hostinger
Continuous delivery target for the web app.

Google Play Console
Android distribution and release tracks.

Google Drive
Final demo recording, shared with link access.

05

SECTION FIVE

Live demo

The next opportunity is the start — and the invoice is the proof.



CLIVORA PREMIUM CAMPAIGN V2

SECTION 05 · LIVE DEMO

The loop I am about to run

One continuous path from a posted job to a paid invoice, across two real accounts.



Two real accounts

A freelancer account and a separate client account, both signed in. Nothing is faked with a demo flag.

Both clients

Part of the loop runs in the browser on clivora.io and part on the Android build, against the same rows.

What I will not claim

Payment is recorded, not processed. There is no live Stripe or escrow, and I will say so on screen.

WHERE THIS LANDED

Results and the evidence behind each claim

8	2	3	$0
Automated suites running in CI	Native clients on one shared record	Sprints, planned and demonstrated	Platform fee on a Connect deal

DELIVERED IN THE INCREMENT

- Repository a grader can navigate: design and testing doc, stories, sprint plan
- CI enforcing web tests, not just building the app
- Ask AI grounded in the user's own CRM, with a written eval set
- A hire-to-invoice loop that runs from a committed script

OPEN, AND TRACKED HONESTLY

- QA-001: web end-to-end journeys still need Playwright coverage
- Outbox behaviour under process death and account switch is untested
- match_crm_embeddings SQL is in the repo, not yet applied in production
- Payments are recorded, not processed — no live gateway

SUBMISSION

What goes into Quantic, and in which of the two buttons

The most common way to lose marks here is uploading the deck into the presentation slot.

BUTTON 1 · PROJECT

Submit the working software

- Live product: https://clivora.io
- Repository: https://github.com/UsmanGhias/clivora
- Agile board: Issues filtered by the capstone label
- Design and testing: docs/capstone/DESIGN_AND_TESTING.md

BUTTON 2 · PRESENTATION

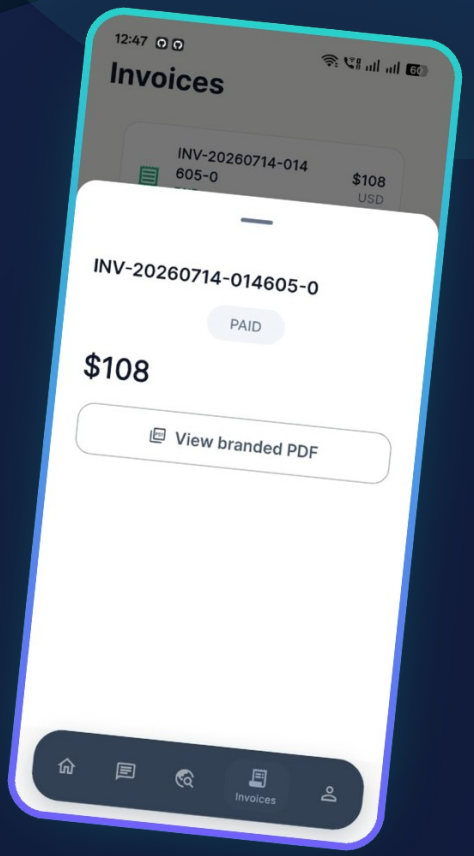
Submit the video, not the slides

- 15–20 minute recording, photo ID shown on camera
- Uploaded to Google Drive with link access for anyone
- Group Project Agreement attached
- Share the repository with the grader account

Do not upload this PPTX, its PDF or a ZIP as the presentation. The presentation submission is the video link. This deck is a speaking aid only.

THANK YOU

Discover the work.
Do the work.
Own the momentum.



Usman Ghias · Product Owner, Scrum Master and engineer · CODCrafters
Happy to take questions on the architecture, the sprint record or the open risks.

LIVE	https://clivora.io
REPOSITORY	https://github.com/UsmanGhias/clivora
DESIGN + TESTING	docs/capstone/DESIGN_AND_TESTING.md

APPENDIX · NOT PRESENTED

Roadmap after the Capstone

Ranked by the risk it removes, not by how impressive it demos.

NEXT

Playwright end-to-end suite

Closes QA-001: login, Connect accept, invoice share.

NEXT

Outbox resilience tests

Process death and account switch under Flutter integration tests.

THEN

Apply match_crm_embeddings

Vector search in a maintenance window, behind a flag.

THEN

Ask AI citations in the UI

Show the invoice or client each sentence came from.

LATER

Payments

A real gateway, only when it can be operated and supported properly.

LATER

iOS

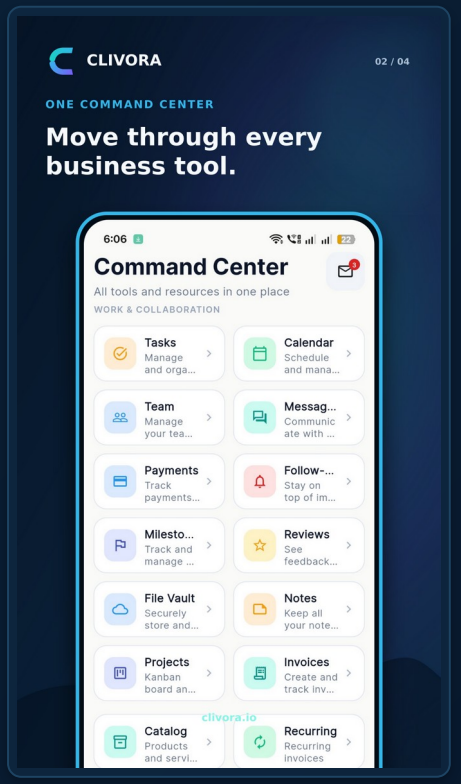
Same Flutter codebase, once Android retention justifies the review overhead.

APPENDIX · NOT PRESENTED

Go-to-market artefacts already produced

Included because the handbook rewards a product, not only a repository.





Premium Campaign V2	Five editorial creatives, a 20-second vertical film and an original 102 BPM soundtrack.
Marketing launch kit	65 assets: square, portrait, story and landscape adaptations, paid-ad variants, WhatsApp statuses.
Google Play listing	Feature graphic and four store screenshots, all generated from real product screens.
Keyword research	126 ranked search terms across freelancer, CRM, invoicing and project-management intent.
Publishing calendar	A seven-day launch sequence with per-platform captions and a fixed posting order.

APPENDIX · PRE-RECORDING CHECKLIST

Run this list before you hit record

Before recording

- Photo ID physically in hand
- 1920x1080 screen, notifications silenced
- Freelancer and client accounts both signed in
- Demo data reset per DEMO_SCRIPT.md
- This deck open in presenter view with notes

During recording

- Show ID and say your full name in the first minute
- Say the word baseline when showing pre-Capstone features
- Never claim Stripe or escrow is live
- Run the whole loop without cutting the recording
- Stay inside 20 minutes

After recording

- Upload to Google Drive, anyone-with-link access
- Test the link in a private browser window
- Attach the Group Project Agreement
- Give the grader account read access to the repo
- Submit both buttons — project and presentation